

A Packing of 10 Four-Spaces in $\mathbb{R}^{10}$

Limin Ge*

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Abstract

We give 10 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.666373. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.639721, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 10 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.666373 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 10$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.639721, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 10/9 = 2.666667$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N10.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
266637381476214200847209342615355184519832176270148744473398349745746763
699309284361096061739642678435385676779661758289348570227805855970546764
6898859994550504578153530048103789388145532307344
```

denominator:

```
999999999997167770987013650521643420846930166808731751582898676369067421
097765268855468981240296775574683057696974567457982540852997670590494784
376370567137392502058622498917758918344126191551
```

B Coordinates

10 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N10.txt`.

```
-397411530722 -700068176272 9482037364 303354739008 104194217491 -189740531625 477068811009 -309451008857 443168897 -50661119897 278882995280 -706597503087
-576469640916 271012701826 208675385203 -387759884969 766198900 -14322275946 388247439233 -268435014003 17648508303 599659017322 -188614445818 -71408565425
65062437397 329377418595 -463514443076 360930816340 21150634230 -704876419520 -342915324191 -121910303075 27648936 -209996146779 270893173497 56852244585
-190310060573 -65093044546 -84028035134 427489523000 92911474810 -328424880992 -330988716854 -324105938825 172522 94636660216 133786864128 573068022963
38911567042 -94419862432 613446233079 276095493743 -338044266865 17053837844 176434822387 -5338218318 18241159245 351457929006 -375504500800 27549288836
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41602242945 -44977991180 -362685416955 -79231519999 1520576297 50610117756 360543137873 -276990835789 -84348046438 268928187139 520116014877 442806795461
-168755511357 147144500720 61073466786 -8798124501 974249109107 -725079620944 63278069283 348554215248 355102001884 -358271451157 -130450090103 337800101418
-497521678769 -87101775937 -129232416120 -401599932390 3317912 42041858055 -557183052437 -75156147597 -385440490335 -619930707320 91987133345 133586655135
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-450657282497 261811264950 -377633433175 -42118657563 162638020 -252452928050 397994544641 -67112473338 70139527 11312119731 -206832831055 -262778035779
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179714986532 30461568844 -154537849674 -579360572336 6939649096 469214443158 27980943646 163622720055 33973797031 333174653238 -42237238130 267604410406
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64308607946 402294903781 270662665409 69325494048 -421894796034 -226609862975 99205056741 -74926174783 49830960322 186928017551 6889824825 -544884012276
-10712173295 269610689420 657059686953 -4654456678 78267166862 -81171529543 576236472317 -102049921043 576220011204 -306383598302 269070367290 -295768132076
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```